KnotBench: A Leakage-Free Benchmark for Machine Learning on Knot Diagrams

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Abstract

Machine learning on knot diagrams is increasingly used to predict knot invariants and to decide knot equivalence, yet reported gains are rarely measured against exact simplification, classical invariants, or tree models given the same inputs. We introduce KnotBench, built entirely from KnotInfo, with knot-level splits by crossing number (train $c \leq 10$, validate $c=11$, test on 2,176 knots with $c=12$), two presentations per knot, size-matched pairs, hard geometric negatives, and controls that isolate what learning adds. Four parameter-matched graph backbones (signed MPNN, GIN, GPS, and a node-marking subgraph GNN) beat diagram-only tree ensembles on unknotting number and smooth 4-genus, but none beats a two-parameter $[c, w]$ regression on signature or Seifert genus at 250 training knots; that gap closes at 636 training knots. On hard negatives drawn from distinct presentations, trained encoders reach only 0.45–0.54 AUROC while an exact hyperbolic certificate scores 1.000 at 1.3 ms per pair. We reproduce the natural-slope result of Davies et al. and document five evaluation pitfalls with measured effect sizes. Data, code and a one-command protocol are released.

1 Introduction

[TODO: prose]

Contributions.

1. **Benchmark.** Knot-level splits by crossing number from KnotInfo; verification across two presentations with size-matched and hard geometric negatives; invariant prediction with a held-out validation slice and no test-set tuning.
2. **Fair and oracle baselines.** Diagram-only tree ensembles see the same information as the graph models; oracle rows that read classical invariants (σ, det, Alexander) are reported separately.
3. **Matched backbones and controls.** Four backbones at $\approx$110k parameters; untrained and shuffled-label controls; paired, Holm-corrected tests for every claimed difference.
4. **Pitfall taxonomy and a reproduction** of the Davies et al. signature result inside the leakage-free split.

Code and data: [TODO: anonymous link].

2 Background and Related Work

Knot diagrams and invariants. PD codes, Reidemeister moves R_1–R_3, flypes [5]; targets σ, g_3, g_4, u.

Machine learning for knots. Davies et al. [3] related signature to cusp geometry, later proved as a natural-slope inequality [2]. [TODO: Hughes; Jejjala et al.; Gukov et al.]

GNN expressivity. 1-WL bound [7]; subgraph GNNs [1]; graph transformers [6].

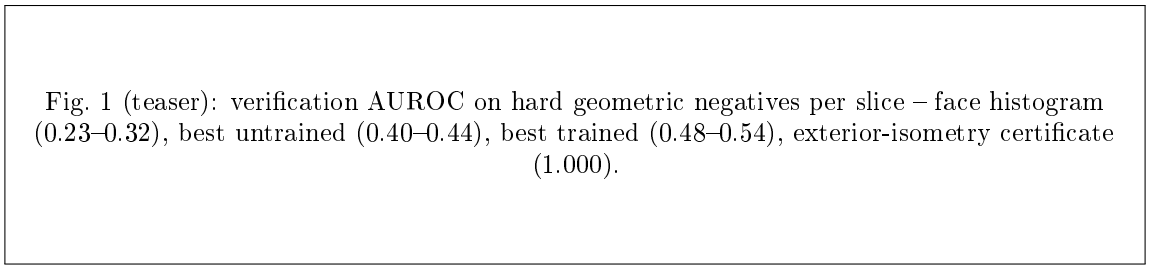


Figure 1: On distinct presentations with hard negatives, learned encoders stay near chance while an exact certificate is perfect and fast. Data: `results/benchmark_v2/task1_hard_negatives.json`.

Benchmark leakage. [TODO:]

3 Benchmark Design

3.1 Data and splits

All 2,978 knots up to 12 crossings and their labels come from KnotInfo [4]. Invariant prediction trains on $c \leq 10$ (250 knots), selects checkpoints per target on $c=11$ (552) and tests on $c=12$ (2,176; $u(K)$ on the 1,628 with an exact value).

3.2 Verification pairs

Both presentations of a knot (PD code and braid closure) come from KnotInfo, which avoids the Perko-pair renumbering at 10_{161}–10_{165}. Diagrams are R_1/R_2-reduced and relabelled. Hard negatives are the two nearest neighbours in face-degree histogram space $[c, n_+, n_-, w, h_2, \ldots, h_{10}]$. Encoders train on all knots with $3 \leq c \leq 8$ plus alternating knots with $c \in \{9, 10\}$ (199 knots), knot-disjoint from the non-alternating test knots.

3.3 Metrics and statistics

AUROC and MAE, mean $\pm$ std over seeds $\{42, \ldots, 46\}$. Differences are tested with paired knot-level bootstrap and Holm correction over all 52 comparisons (Appendix B).

4 Models, Baselines and Controls

Proposition 1 (Confluent R_1/R_2 reduction)**.** *Exhaustive R_1-curl and R_2-bigon removal terminates in $O(|V|)$ steps and its result is independent of move order. [TODO: state precisely; proof in Appendix A]*

- **Backbones** (hidden width chosen so each has $\approx$110k parameters; 4 layers; AdamW, 60 epochs, batch 32): signed MPNN, GINE [7], GPS [6], DS-GNN with node marking [1].
- **Diagram-only trees**: RF and GBDT on the 13 diagram features above.
- **Oracle trees** (not competitors): diagram features plus σ, $|\sigma|/2$, det, Alexander coefficients; σ columns removed when σ is the target.
- **Classical**: WL hash, $|\sigma|/2$ bound, $\deg \Delta/2$, exterior isometry certificate (ceiling).

5 Results

Graph models beat diagram-only trees, not classical invariants. On $u(K)$ and $g_4(K)$ every backbone except GPS on u beats diagram-only RF (e.g. GIN -0.035 MAE on u, -0.034 on g_4; $p_{\text{Holm}}=0.010$). Trees given σ reach 0.137 on g_4, so the earlier "RF beats GNN" result came from oracle inputs.

Table 1: Invariant prediction on unseen c=12 knots (train $c\leq$10, checkpoint on c=11), MAE, mean ± std over 5 seeds. Bold: best non-oracle entry. Grey rows read classical invariants and are not competitors.

		Trivial		Diagram-only trees		Graph backbones (param.-matched)			
Target	N	Mean	Lin. $[c, w]$	RF	GBDT	Signed MPNN	GIN	GPS	Subgraph (DS-GNN)
$u(K)$	1628	0.489	0.471	0.432 ± 0.003	0.426 ± 0.003	0.396 ± 0.013	0.408 ± 0.017	0.436 ± 0.018	0.401 ± 0.008
$g_4(K)$	2176	0.597	0.524	0.388 ± 0.009	0.385 ± 0.004	0.363 ± 0.018	0.361 ± 0.008	0.366 ± 0.011	0.364 ± 0.022
$g_3(K)$	2176	0.855	**0.619**	0.744 ± 0.003	0.723 ± 0.014	0.729 ± 0.022	0.669 ± 0.024	0.747 ± 0.038	0.693 ± 0.074
$\sigma(K)$	2176	2.325	**0.921**	0.989 ± 0.007	0.974 ± 0.011	1.004 ± 0.041	0.947 ± 0.020	1.008 ± 0.044	0.974 ± 0.048
Oracle inputs (not competitors): trees given σ, det, *Alexander coefficients; classical formulas*									
$u(K)$		RF 0.372 ± 0.011, GBDT 0.373 ± 0.011; closed form 0.848							
$g_4(K)$		RF 0.137 ± 0.006, GBDT 0.310 ± 0.029; closed form 0.271							
$g_3(K)$		RF 0.056 ± 0.001, GBDT 0.056 ± 0.000 (formula input); closed form 0.015							
$\sigma(K)$		RF 0.857 ± 0.005, GBDT 0.858 ± 0.008 (σ columns removed)							

Table 2: Verification across distinct presentations (AUROC, mean ± std over 5 seeds). Ceiling = exterior isometry certificate. Backbone columns show the better of supervised and contrastive training; Untrained is the best of the four untrained backbones.

Slice	Neg.	Ceiling	Face hist.	Untrained	Signed MPNN	GIN	GPS	Subgraph (DS-GNN)
$c \leq 10$ (50)	random	1.000	0.937	0.675 ± 0.017	0.642 ± 0.040	0.840 ± 0.097	0.656 ± 0.060	0.656 ± 0.048
$c \leq 10$ (50)	hard	1.000	0.321	0.428 ± 0.024	0.453 ± 0.034	0.483 ± 0.040	0.486 ± 0.032	0.483 ± 0.035
$c = 11$ (185)	random	1.000	0.867	0.677 ± 0.005	0.656 ± 0.022	0.802 ± 0.074	0.672 ± 0.035	0.693 ± 0.025
$c = 11$ (185)	hard	1.000	0.230	0.401 ± 0.007	0.477 ± 0.034	0.507 ± 0.030	0.493 ± 0.022	0.479 ± 0.032
$c = 12$ (100)	random	1.000	0.909	0.697 ± 0.012	0.750 ± 0.045	0.834 ± 0.076	0.742 ± 0.039	0.720 ± 0.016
$c = 12$ (100)	hard	1.000	0.267	0.441 ± 0.015	0.506 ± 0.023	0.543 ± 0.059	0.492 ± 0.078	0.481 ± 0.013

Signature and Seifert genus are data-limited. At 250 training knots no diagram model beats linear $[c, w]$ on σ (GIN +0.007, n.s.) or g_3. With 636 training knots GIN reaches 0.718 on σ and 0.472 on g_3 versus 0.919 and 0.613 for $[c, w]$ (Table 3).

Learning adds little on hard negatives. Training lifts the best encoder from 0.40–0.44 to 0.48–0.54 AUROC (significant at c=11, 12, not at $c\leq$10); all remain far below the certificate.

Cost. The certificate costs 1.3 ms per pair (median), comparable to a single encoder forward pass (Table 5).

Reproducing Davies et al. Hyperbolic invariants predict σ far better than any diagram model (Table 4).

6 A Taxonomy of Pitfalls

7 Relevance to Scientific Practice

KnotBench answers a practical question for working scientists: when should a learned model replace or complement an exact method? For knot equivalence it should not: an exterior-isometry certificate is correct on every test pair at 1.3 ms per pair, while trained encoders stay near chance on hard negatives. For invariants without fast algorithms—unknotting number and smooth 4-genus—all four graph backbones carry significant signal beyond diagram-only baselines, which makes them useful for ranking open cases (548 of 2,176 twelve-crossing knots lack an exact $u(K)$ in KnotInfo) but not for settling them. The same protocol applies to knotted proteins, circular DNA and polymer simulations, where knot identification runs at database scale: screen with exact invariants, and adopt a learned model only if it beats them on accuracy and cost under a knot-disjoint split. The oracle-input pitfall (g_4 MAE 0.388 → 0.137 once σ, which helps define the labels, is an input) is a general hazard for scientific benchmarks whose labels come from bounds.

Table 3: Learning curve: test MAE on c=12 as the training set grows (validation fixed at 30% of c=11).

Target	Model	c≤9 (85)	c≤10 (250)	c≤10+70%c11 (636)
$u(K)$	Lin. $[c, w]$	0.540 ± 0.000	0.471 ± 0.000	0.445 ± 0.000
	RF (diagram)	0.477 ± 0.009	0.432 ± 0.003	0.423 ± 0.002
	Signed MPNN	0.450 ± 0.010	0.408 ± 0.007	0.404 ± 0.014
	GIN	0.469 ± 0.035	0.409 ± 0.015	0.388 ± 0.007
	GPS	0.436 ± 0.018	0.441 ± 0.021	0.427 ± 0.007
	Subgraph (DS-GNN)	0.448 ± 0.016	0.420 ± 0.017	0.427 ± 0.007
$g_4(K)$	Lin. $[c, w]$	0.575 ± 0.000	0.524 ± 0.000	0.514 ± 0.000
	RF (diagram)	0.441 ± 0.008	0.388 ± 0.009	0.385 ± 0.003
	Signed MPNN	0.437 ± 0.003	0.359 ± 0.012	0.330 ± 0.009
	GIN	0.419 ± 0.010	0.361 ± 0.008	0.314 ± 0.009
	GPS	0.432 ± 0.022	0.366 ± 0.011	0.347 ± 0.012
	Subgraph (DS-GNN)	0.431 ± 0.012	0.362 ± 0.020	0.342 ± 0.019
$g_3(K)$	Lin. $[c, w]$	0.594 ± 0.000	0.619 ± 0.000	0.613 ± 0.000
	RF (diagram)	1.036 ± 0.008	0.744 ± 0.003	0.613 ± 0.002
	Signed MPNN	0.963 ± 0.070	0.729 ± 0.022	0.574 ± 0.022
	GIN	0.912 ± 0.077	0.672 ± 0.030	0.472 ± 0.028
	GPS	1.024 ± 0.031	0.743 ± 0.037	0.586 ± 0.012
	Subgraph (DS-GNN)	0.978 ± 0.055	0.693 ± 0.073	0.564 ± 0.024
$\sigma(K)$	Lin. $[c, w]$	0.896 ± 0.000	0.921 ± 0.000	0.919 ± 0.000
	RF (diagram)	1.180 ± 0.013	0.989 ± 0.007	0.966 ± 0.003
	Signed MPNN	1.347 ± 0.040	1.013 ± 0.042	0.850 ± 0.020
	GIN	1.211 ± 0.026	0.950 ± 0.021	0.718 ± 0.052
	GPS	1.342 ± 0.068	1.007 ± 0.044	0.918 ± 0.058
	Subgraph (DS-GNN)	1.412 ± 0.055	0.971 ± 0.051	0.899 ± 0.058

8 Reproducibility Statement

All headline numbers are produced on CPU with seeds $\{42, \ldots, 46\}$ by one script and written to JSON, from which every table is generated. Repeated runs, different hash seeds and thread counts give bit-identical test predictions (identical SHA-256 over all 2,176 predictions). The full suite takes about 2 CPU-hours with no GPU. One earlier value was platform-dependent: the learned gain in the c≤8 protocol was +0.008 AUROC on Apple MPS with unseeded initialisation and is +0.027 [+0.013, +0.042] on seeded CPU runs; we report the latter. We release pinned dependencies (SnapPy, spherogram, scikit-learn, database_knotinfo), data with checksums, and record the commit, package versions and data hash in every results file. [TODO: confirm once the lockfile, Dockerfile and DOI exist; add cross-platform tolerance]

9 Limitations

Crossing numbers ≤ 12; synthetic perturbations; KnotInfo labels for u and g_4 exist where classical bounds meet, which biases the solved subset; R_3 is not canonicalized; 5 of 1,336 certificate pairs are non-hyperbolic and use knot Floer homology equality instead; the learned gain in the c≤8 protocol (+0.027 AUROC on CPU) differs from the earlier MPS run (+0.008).

10 Conclusion

[TODO: two sentences + how to submit results to the benchmark]

Table 4: Signature from hyperbolic invariants of the complement (Davies et al. setting vs. the KnotBench crossing split); 5 seeds; "exact" rounds to the nearest even integer.

	Random 80/20 (Davies et al. setting)		KnotBench $c \leq 10 \to c = 12$	
Predictor of $\sigma(K)$	MAE	Exact acc.	MAE	Exact acc.
Mean	2.292	–	2.316	–
Natural slope / 2 (formula)	0.713	0.739	0.686	0.739
MLP (hyperbolic inv.)	0.501 ± 0.013	0.862	0.596 ± 0.011	0.831
GBDT (hyperbolic inv.)	0.424 ± 0.003	0.873	0.542 ± 0.005	0.808
RF (hyperbolic inv.)	0.344 ± 0.002	0.868	0.476 ± 0.002	0.804

Table 5: Runtime on one CPU core.

Method	Params	Time per diagram / pair
Exterior isometry (certificate)	–	1.3 ms (median), 2.1 ms (p95)
WL hash	–	0.71 ms
Signed MPNN encoder	123,207	0.57 ms
GIN encoder	119,555	0.65 ms
GPS encoder	131,535	0.54 ms
Subgraph (DS-GNN) encoder	123,271	4.30 ms

References

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[7] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations*, 2019.

A Proof of Confluent R_1/R_2 Reduction

[TODO: from docs/reidemeister_invariance_theory.md]

B Statistical Tests

Paired knot-level bootstrap; Δ is MAE(A)−MAE(B) for Task 2 and AUROC(A)−AUROC(B) for Task 1, computed on seed-averaged predictions; Holm correction over the full family. Resampling floors p at 1/reps.

Table 6: Pitfalls, how to detect them, and measured effect.

Pitfall	Detection / fix	Measured effect
Traversal-order leakage	Cyclic relabel + orientation reversal; face-degree features	Chord AUROC 0.918 → 0.401
Diagram-size imbalance	Match step schedules for positives/negatives	Spurious 0.000 → 0.512 AUROC
Encoder as graph hash	Untrained and shuffled-label controls; hard negatives	Untrained 0.95 on easy pairs, 0.40 on hard
Oracle / label leakage	Audit features against label definitions	σ MAE 0.010 (leak); g_4 0.388 → 0.137 with σ
Multi-task checkpointing	Per-target selection on validation	Joint vs. per-target selection

Comparison (A vs B)	Δ	95% CI	p	p_{Holm}
gain / canonical_r1r2 vs random_canon	+0.027	[+0.013, +0.042]	0.0005	0.010
gain / canonical_r1r2 vs shuffled_canon	+0.029	[+0.015, +0.045]	0.0005	0.010
gain / canonical_r1r2 vs fixed_chord_canon	+0.058	[+0.038, +0.079]	0.0005	0.010
task1 / c≤10_hard_geometric / gin_contrastive vs gin_untrained	+0.072	[-0.029, +0.167]	0.1680	1.000
task1 / c≤10_hard_geometric / gin_contrastive vs face_histogram	+0.175	[+0.073, +0.270]	0.0005	0.010
task1 / c=11_hard_geometric / gin_supervised vs gin_untrained	+0.106	[+0.074, +0.139]	0.0005	0.010
task1 / c=11_hard_geometric / gin_supervised vs face_histogram	+0.268	[+0.224, +0.311]	0.0005	0.010
task1 / c=12_hard_geometric / gin_contrastive vs gin_untrained	+0.135	[+0.069, +0.202]	0.0005	0.010
task1 / c=12_hard_geometric / gin_contrastive vs face_histogram	+0.309	[+0.237, +0.380]	0.0005	0.010
task2 / unknotting_number / gnn_mpnn vs rf_diagram	-0.041	[-0.057, -0.025]	0.0002	0.010
task2 / unknotting_number / gnn_gin vs rf_diagram	-0.035	[-0.051, -0.019]	0.0002	0.010
task2 / unknotting_number / gnn_gps vs rf_diagram	-0.006	[-0.022, +0.011]	0.5028	1.000
task2 / unknotting_number / gnn_subgraph vs rf_diagram	-0.038	[-0.054, -0.021]	0.0002	0.010
task2 / unknotting_number / gnn_mpnn vs linear_cw	-0.081	[-0.094, -0.069]	0.0002	0.010
task2 / unknotting_number / gnn_gin vs linear_cw	-0.075	[-0.088, -0.062]	0.0002	0.010
task2 / unknotting_number / gnn_gps vs linear_cw	-0.046	[-0.061, -0.031]	0.0002	0.010
task2 / unknotting_number / gnn_subgraph vs linear_cw	-0.078	[-0.091, -0.065]	0.0002	0.010
task2 / unknotting_number / rf_oracle vs rf_diagram	-0.060	[-0.071, -0.050]	0.0002	0.010
task2 / unknotting_number / rf_diagram vs linear_cw	-0.040	[-0.057, -0.023]	0.0002	0.010
task2 / smooth_four_genus / gnn_mpnn vs rf_diagram	-0.037	[-0.051, -0.023]	0.0002	0.010
task2 / smooth_four_genus / gnn_gin vs rf_diagram	-0.034	[-0.047, -0.020]	0.0002	0.010
task2 / smooth_four_genus / gnn_gps vs rf_diagram	-0.027	[-0.040, -0.013]	0.0002	0.010
task2 / smooth_four_genus / gnn_subgraph vs rf_diagram	-0.042	[-0.055, -0.028]	0.0002	0.010
task2 / smooth_four_genus / gnn_mpnn vs linear_cw	-0.175	[-0.188, -0.162]	0.0002	0.010
task2 / smooth_four_genus / gnn_gin vs linear_cw	-0.171	[-0.186, -0.156]	0.0002	0.010
task2 / smooth_four_genus / gnn_gps vs linear_cw	-0.165	[-0.178, -0.151]	0.0002	0.010
task2 / smooth_four_genus / gnn_subgraph vs linear_cw	-0.179	[-0.193, -0.166]	0.0002	0.010
task2 / smooth_four_genus / gnn_subgraph vs gnn_mpnn	-0.005	[-0.007, -0.002]	0.0008	0.010
task2 / smooth_four_genus / rf_oracle vs rf_diagram	-0.250	[-0.265, -0.235]	0.0002	0.010
task2 / smooth_four_genus / rf_diagram vs linear_cw	-0.138	[-0.156, -0.119]	0.0002	0.010
task2 / three_genus / gnn_mpnn vs rf_diagram	-0.029	[-0.043, -0.015]	0.0002	0.010
task2 / three_genus / gnn_gin vs rf_diagram	-0.086	[-0.101, -0.070]	0.0002	0.010
task2 / three_genus / gnn_gps vs rf_diagram	-0.002	[-0.015, +0.013]	0.8184	1.000
task2 / three_genus / gnn_subgraph vs rf_diagram	-0.079	[-0.093, -0.066]	0.0002	0.010
task2 / three_genus / gnn_mpnn vs linear_cw	+0.095	[+0.075, +0.116]	0.0002	0.010
task2 / three_genus / gnn_gin vs linear_cw	+0.039	[+0.020, +0.058]	0.0002	0.010
task2 / three_genus / gnn_gps vs linear_cw	+0.123	[+0.102, +0.143]	0.0002	0.010
task2 / three_genus / gnn_subgraph vs linear_cw	+0.045	[+0.026, +0.065]	0.0002	0.010
task2 / three_genus / gnn_gin vs gnn_mpnn	-0.057	[-0.062, -0.051]	0.0002	0.010
task2 / three_genus / rf_oracle vs rf_diagram	-0.687	[-0.709, -0.666]	0.0002	0.010
task2 / three_genus / rf_diagram vs linear_cw	+0.124	[+0.105, +0.144]	0.0002	0.010
task2 / signature / gnn_mpnn vs rf_diagram	-0.022	[-0.053, +0.008]	0.1488	1.000

Comparison (A vs B)	Δ	95% CI	p	p_{Holm}
task2 / signature / gnn_gin vs rf_diagram	-0.059	[-0.090, -0.030]	0.0002	0.010
task2 / signature / gnn_gps vs rf_diagram	-0.006	[-0.035, +0.024]	0.7384	1.000
task2 / signature / gnn_subgraph vs rf_diagram	-0.058	[-0.088, -0.030]	0.0002	0.010
task2 / signature / gnn_mpnn vs linear_cw	+0.043	[+0.019, +0.068]	0.0012	0.010
task2 / signature / gnn_gin vs linear_cw	+0.006	[-0.020, +0.032]	0.6304	1.000
task2 / signature / gnn_gps vs linear_cw	+0.060	[+0.034, +0.086]	0.0002	0.010
task2 / signature / gnn_subgraph vs linear_cw	+0.007	[-0.016, +0.031]	0.5432	1.000
task2 / signature / gnn_gin vs gnn_mpnn	-0.037	[-0.054, -0.020]	0.0002	0.010
task2 / signature / rf_oracle vs rf_diagram	-0.134	[-0.162, -0.108]	0.0002	0.010
task2 / signature / rf_diagram vs linear_cw	+0.066	[+0.040, +0.092]	0.0002	0.010

C Secondary Protocol (train $c \leq 11$, no validation)

Reported in `results/rigorous_tasks_benchmark.json` (`task2_invariants`); superseded by Table 1.

D Hyperparameters and Selection Rules

[TODO:]

E Compute

[TODO: hardware, CPU-hours, per-run time]

F Datasheet for KnotBench

[TODO: motivation, composition, collection, uses, distribution, maintenance; Croissant metadata]

NeurIPS Paper Checklist

[TODO: official checklist]